



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Bayesian Networks
- **Course Outcome:** CO2
- **Activity Chosen:** Think-Pair- Share
- **Justification:**

Bayesian networks were chosen for the Think-Pair-Share activity to promote active learning and peer collaboration. The concept involves modeling uncertain relationships using directed acyclic graphs, which can be complex for students to grasp individually. Working in pairs allows students to share insights and troubleshoot difficulties in constructing and reasoning about probabilistic models. The activity encourages critical thinking as students explore conditional dependencies and probabilistic inference. Through peer discussions, students solidify their understanding and refine their models. Bayesian networks are applicable in fields like AI, medicine, and economics, making the topic relevant and engaging. The collaborative approach also fosters problem-solving skills and enhances real-world application. It encourages both theoretical understanding and practical skills. This method helps students move from passive to active learners. Ultimately, it strengthens their ability to reason under uncertainty and make informed decisions.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - ✓ Instructor explains Bayesian networks and students review related materials before class.
 - ✓ **Think (5 Minutes):** Students individually consider how to represent a real-world problem using a Bayesian network.
 - ✓ **Pair (5 Minutes):** Students pair up to share their ideas, compare network structures, and discuss dependencies and challenges.

- ✓ **Share (Class Discussion):** Pairs share their networks with the class; the instructor clarifies concepts and answers questions.
- ✓ Finally share points with the entire class (Share) as shown in Fig 1 & 2
- ✓ **Outcome:** This activity helps students reinforce their understanding of Bayesian networks, conditional dependencies, and probabilistic reasoning through individual reflection and peer discussion.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	2
Justification for correlation	Requires basic engineering knowledge to implement Bayesian Network	Able to identify engineering problems with Bayesian Network	Able to design solutions for complex engineering problems using Bayesian Network	Able to solve problem individually or in team and using Bayesian Network	Able to explain the way to solve the problem using Bayesian Network with their team members.	Able to apply Bayesian Networks to solve some computational problem.

• **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members [Divya Sri J.S, Gayathri.K, Pavithra. S, Poornima. A]



Figure 2: Solved answers shared by the team members [Poongothai. S, Malarkodi. R, Jeyadevi. S, Meenadevi.M]

• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- All students felt that they engaged fully in the task and discussions.
- Later Students informed that they were explored and shared Bayesian Networks concepts with peers.
- Students also informed that pairing up allowed them to learn from each other and clarify their doubts.

❖ ***Benefit of the practice:***

- The activity helped students feel more confident in explaining Bayesian Network concept.
- From this activity, the students got clear idea about how to apply Bayesian Network to practical scenarios.

❖ ***Challenges faced in implementation:***

- Some students were less active in the discussions, which impacted the overall group dynamics.
- Few students struggled to communicate their ideas clearly, causing confusion within their teams.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf
- ❖ <https://www.theteachertoolkit.com/index.php/tool/think-pair-share>
https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VI_Think_Pair_and_Share.pdf

Signature of Faculty Member

HOD